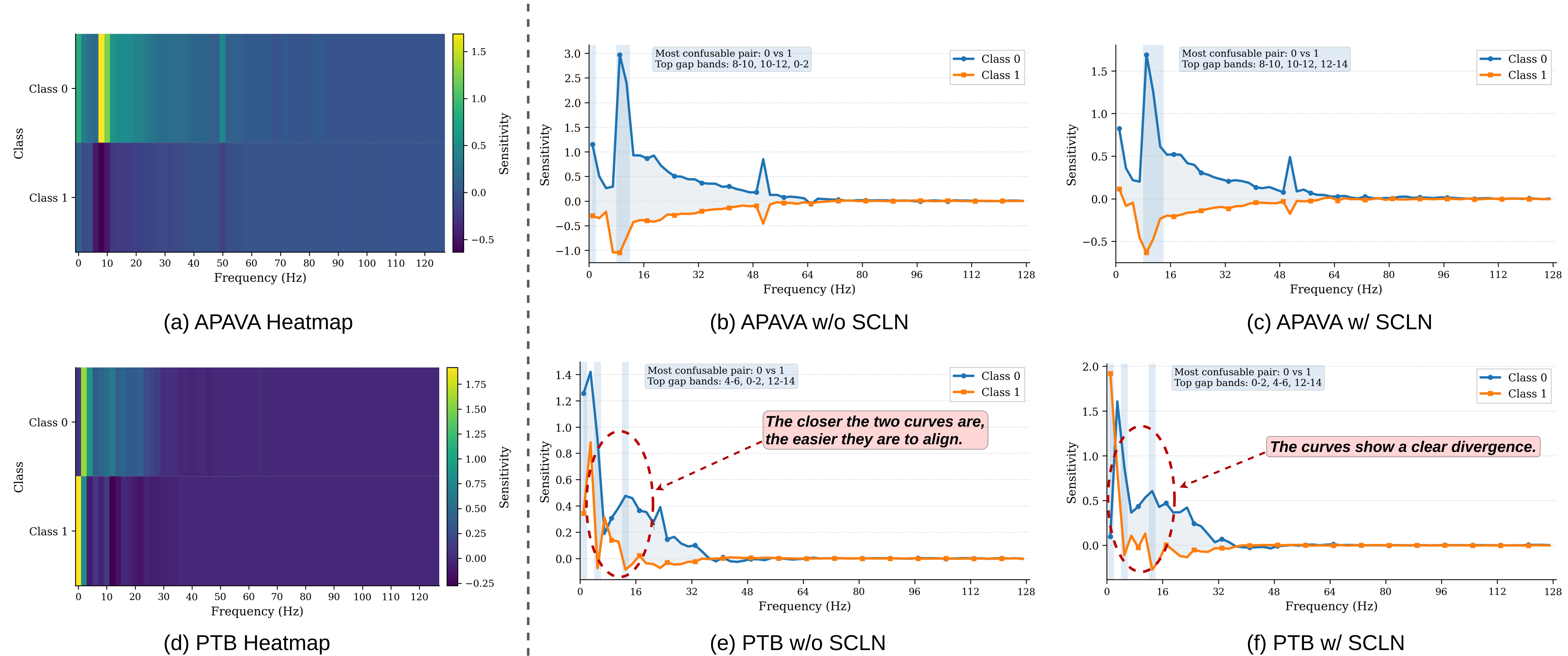


$$\text{ sensitivity : } s_i(b) = f_{y_i}(x_i) - f_{y_i}(x_i^{-b})$$

Here $f_{y_i}(x_i)$ denotes the model score (logit or probability) for the true class of sample x_i , and x_i^{-b} represents the signal with the b -th frequency band removed. The sensitivity $s_i(b)$ measures how much the model's confidence in the correct class decreases when a specific frequency band is masked. A larger value indicates that the band is more important for correctly recognizing that class.



The figure presents a band-wise sensitivity analysis on both APAVA and PTB datasets, including heatmaps and sensitivity curves for the most confusable class pair. The heatmaps ((a) and (d)) illustrate how different frequency bands contribute to each class, revealing that discriminative information is concentrated in low-frequency regions. The curve plots ((b), (c), (e), and (f)) further compare class-wise sensitivities before and after applying SCLN. Without SCLN, the sensitivity curves of the two classes are relatively close in key frequency bands, indicating that their representations are more easily aligned. In contrast, with SCLN, the curves exhibit clearer divergence, suggesting that the model captures more class-specific frequency characteristics and enhances discriminability between classes. Overall, the results demonstrate that SCLN improves the separation of class-dependent spectral features.